HADGSM: A Unified Nonconvex Framework for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection aims at distinguishing targets of interest from the background without prior knowledge. Although low-rank representation (LRR)-based methods have been broadly applied in anomaly detection tasks, how to approximate the penalties in LRR-based methods more precisely is still a problem that needs to be further investigated. To this end, this article designs a unified nonconvex framework called hyperspectral anomaly detection via generalized shrinkage mappings (HADGSMs) to better approximate the LRR-based methods. The core of the proposed framework is to design new nonconvex penalties to approximate the group sparsity, l_0 gradient, and low-rankness penalties in the LRR-based anomaly detection models, which can be efficiently minimized by means of generalized shrinkage mappings (GSMs). Then, an efficient alternating direction method of multipliers (ADMM) is developed to handle the proposed model. Experiments conducted on several real hyperspectral datasets demonstrate the superiority and effectiveness of the proposed framework in enhancing detection performance with respect to state-of-the-art methods.**

***Index Terms*— Anomaly detection, generalized shrinkage mappings (GSMs), hyperspectral image, low-rank representation (LRR), total variation (TV).**

I. Introduction

HYPERSPECTRAL imagery records reflected light in hundreds of narrow bands of ground objects [1], [2]. The abundant spectral information of hyperspectral imagery provided a special diagnostic identification ability, leading to increasing interest in image analysis, such as unmixing [3], [4], [5], [6], denoising [7], [8], dimension reduction [9], [10], superresolution [11], [12], [13], classification [14], [15], [16], and target detection [17]. Among them, target detection is of great significance in hyperspectral image exploitation, which aims at distinguishing targets of interest from the background [18].

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Target detection can be divided into supervised and unsupervised approaches in accordance with the prior information on objects. The supervised-based approaches require that the spectral characteristics of the desired targets are known in advance. In these situations, the spectral characteristics of the targets can be defined by a target subspace or a single target spectrum. The background, on the other hand, can be modeled by a subspace or a Gaussian distribution [19]. Enormous works such as the matched subspace detector (MSD) [20], linear spectral matched filter (SMF) [21], adaptive subspace detector (ASD) [22], orthogonal subspace projection (OSP) [23], sparse representation (SR) [24], sparse and low-rank matrix decomposition (SLRMD) [25], and chessboard topology [26] have been widely investigated in the target detection task and manifested stable detection performance.

The unsupervised-based approaches, also known as anomaly detection, distinguish targets of interest from the background without prior knowledge. In such approaches, pixels with spectral signatures that differ significantly from their neighboring background clutter pixels are identified as anomalies. Many approaches that can be divided into several categories: 1) statistic-based; 2) distance-based; 3) deep learning-based; and 4) reconstruction-based have been proposed to handle the anomaly detection task in the literature. Among them, the Reed–Xiaoli (RX) algorithm based on the assumption that the background obeys a multivariate normal distribution is the most popular statistical approach [27]. A number of statistical-based approaches that further improve the performance of the detection have been presented, including the subspace RX [28], regularized RX [29], and local RX [30]. These types of methods may fail in the case of complex situations where there are many types of background objects and their distribution is irregular. The distance-based methods assume that the pixels are grouped in the light of the distance, and pixels that deviate from the center of the cluster are regarded as anomalies. The support vector machine method [31] and its variant [32], [33] adopt a class classifier to estimate the smallest closed hypersphere supported by the background pixels, and pixels beyond the hypersphere are regarded as anomalies. Furthermore, graph theory that calculates the similarity between two pixels has been widely used in the anomaly detection model [34], [35]. However, the accuracy of targets for these distance-based methods depends on whether the parameters can be appropriately set. Techniques based on deep learning can be divided into supervised or unsupervised deep learning depending on whether the labeled sample is used. The typical supervised

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deep learning method is the classic convolution neural network (CNN) [36] but detection accuracy is restricted by the quantity and quality of data. Furthermore, the model mismatch and training time also limit its application. The representative unsupervised deep learning networks including generative adversarial networks (GANs) [37] and autoencoder (AE) [38] for anomaly detection have been proposed, but they can possibly learn the identity mapping [39], leading to unsatisfactory detection results. Very recently, blind spot networks based on self-supervised learning have been introduced into the anomaly detection tasks for the first time [40], [41], [42], which avoid the above issues and yield promising outcomes.

With the development of machine learning and compressed sensing, reconstruction-based approaches have been developed and achieved enormous progress [18]. The core idea of the reconstruction-based approach is to reconstruct the background pixels using a particular dictionary in a reasonable model, and the residual is regarded as the anomaly pixels. Depending on the restriction on the dictionary, the reconstruction-based anomaly detection approaches can be divided into SR-based, collaborative representation (CR)-based, and low-rank representation (LRR)-based methods. The SR-based method [43] was proposed based on the conception that the normal pixel can be represented by several atoms from the overcomplete dictionary, while anomalies cannot. The CR-based method [44] assumes the background pixels can be represented by their spatial surroundings, achieving through a linear combination of surrounding neighbors of each test pixel. In general, both the SR-based and CR-based approaches detect images pixel-by-pixel, ignoring the global structure of hyperspectral images.

The LRR-based approaches that consider the strong correlation between pixels in hyperspectral images have been broadly applied in anomaly detection. These methods aim at finding the lowest rank representation of the background part under a certain dictionary and the sparsest representation of the anomaly part [45], [46]. The LRR model degenerates into the robust principal component analysis (RPCA) model [47] under the assumption that the pixels of a hyperspectral image are drawn from a single subspace. The RPCA-RX was developed in [48] as the first step to decompose the hyperspectral image into the background part with low-rank property and the anomaly part with the sparse property using RPCA. Then the RX algorithm was used on the sparse part to detect the anomalies. Based on the RPCA, the low-rank and sparse decomposition-based Mahalanobis distance method (LSMAD) was developed in [49] using the Mahalanobis distance for similarity measurement. The low-rank and SR (LRASR) detection method [50] introduced an extra sparse constraint term in the LRR model to better describe the local structure of the representation coefficients. The anomaly detector based on LRR and learned dictionary has been introduced in [51] to further enhance the robustness of regularization parameters. A data-driven projection LRR method has been presented in [52] to realize dimensionality reduction and separate anomalies simultaneously. A recent work in [53] first proposed a t-linear tensor subspace learning model for hyperspectral image analysis, which can simultaneously achieve dimensionality reduction and noise rejection. A spatial constraint imposed on the LRR coefficients with single or multiple local window modes has been developed in [54] to improve detection performance. A modified LRASR that models the sparse component as a mixture of Gaussian has been designed in [55] to handle complex noise distribution. The spectral unmixing and dictionary-based low-rank decomposition method has been developed in [56] by applying spectral unmixing to obtain the abundance maps and then inputting the abundance maps into the LRR model for detection. To account for the spatial information, the graph and total variation (TV) regularizers have been embedded in the LRR model to preserve the spatial relationships and local geometrical structure in hyperspectral images [57]. To alleviate the effect of background interference, the tensor principal component analysis (TPCA)-based detector has been proposed in [58] to combine the preprocessing method with traditional detectors. A prior-based tensor approximation (PTA) method has been proposed in [59], in which truncated nuclear norm and linear TV norm have been embedded in the RPCA model to capture the low-rank prior and piecewise-smooth prior. A tensor LRR-based method has been developed in [60] to achieve better detection performance, in which the hyperspectral image is decomposed into a background tensor and an anomaly tensor. The background part was represented by a tensorial background dictionary and the corresponding low-rank coefficients, while the anomaly part was characterized by the spatial group sparse prior. A unified anomaly detection and noise removal (ADNR) paradigm has been presented in [61], which jointly learns double subspace representation and alternatively updates processing and interpretation tasks. The low-dimensional orthogonal basis was used to represent hyperspectral images and distinctive features were extracted for anomaly detection, and then the complex distinctive features were represented by the dictionary-based low-rank subspace.

Although the above LRR-based methods have achieved remarkable success in anomaly detection tasks, most methods use convex penalties in the models resulting in insufficient expression for the spatial correlation. Recent works in [62] and [63] have derived a class of generalized shrinkage mappings (GSMs) and induced nonconvex penalties that better approximate the l_0 penalty and empirically can enhance the solutions for the unmixing task. Motivated by the outstanding performance in [62], [63], and [64], this article proposes a novel unified framework called hyperspectral anomaly detection via GSMs (HADGSMs) to further improve detection performance. The proposed framework provides a new nonconvex manner to approximate the group sparsity, l_0 gradient, and low-rankness penalties in the LRR-based models using the GSMs [63], [65]. More specifically, the HADGSM model approximates the $l_{2,0}$ penalty using the GSM-induced group sparse penalty to more precisely exact the nonzero columns of outliers and approximates the l_0TV penalty using the GSM-induced TV penalty to further enhance the piecewise smoothness and maintain the sharp edges

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between the background and anomalies and approximates the low-rankness penalty using the GSM-induced nuclear penalty to express the spatial correlation sufficiently. Unlike previous nonconvex-related models with an explicit expression of penalties, the HADGSM method does not have the closed-form expression of penalties but only relevant GSMs. The motivation for choosing the alternative GSMs is to have them as closed-form proximal mappings, which requires them to actually be the proximal mappings of the nonconvex penalty functions. In practice, it is much more useful for most state-of-the-art algorithms to know the explicit proximal mappings rather than the closed-form expression of the induced penalty functions. Compared with the existing LRR-based techniques, the main contributions of this work are summarized as follows.

1) A unified nonconvex framework HADGSM was developed for anomaly detection tasks. Spectral correlation, spatial smoothness, and intrinsic local geometrical structure priors are all incorporated into the proposed framework. The proposed framework contains most existing LRR-based models as special cases.
2) To the best of our knowledge, this is the first time to expand the GSMs in the LRR-based hyperspectral anomaly detection models. Three new GSM-induced penalties were introduced in the HADGSM model, providing a more accurate characterization of the background part and anomaly part in hyperspectral images.
3) The alternating direction method of multipliers (ADMM) was designed to optimize the proposed model. Experiments conducted on real hyperspectral datasets validate the better detection performance of the proposed framework without sacrificing computational efficiency, in comparison to classic LRR-based anomaly detectors.

The rest of this work is organized as follows. Section II reviews the related work. Section III describes the proposed framework. Section IV shows the experimental results on the real hyperspectral datasets. The conclusion is drawn in Section V.

II. Related Work

In this section, we first depict the reconstruction-based model and then review several related state-of-the-art anomaly detection models.

A. Reconstruction-Based Model

The reconstruction-based model is a basic model to handle anomaly detection tasks, which can be written as follows [18]:

$$\mathbf{X} = \mathbf{DS} + \mathbf{E} \tag{1}$$

where $\mathbf{X} := [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{L\times n}$ denotes the observed hyperspectral images with n pixels and each $\mathbf{x}_i \in \mathbb{R}^{L\times 1}$ denotes the L-spectrum measured pixel; $\mathbf{DS}$ denotes the background component; $\mathbf{D} := [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_m] \in \mathbb{R}^{L\times m}$ and $\mathbf{S} := [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_n] \in \mathbb{R}^{m\times n}$ denote the dictionary and corresponding representation coefficients, respectively; and $\mathbf{E} := [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n] \in \mathbb{R}^{L\times n}$ denotes the anomaly component.

B. Low-Rank Representation Model

Based on the reconstruction-based model, the LRR model aims at finding the lowest rank representation of the background component under the given dictionary and the sparsest representation of the anomaly component, which can be expressed as follows [45], [46]:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} \operatorname{rank}(\mathbf{S}) + \lambda \|\mathbf{E}\|_{2,0} \\ &\text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \end{aligned} \tag{2}$$

where $\lambda > 0$ denotes the regularization parameter; $\|\cdot\|_{2,0}$ counts the number of nonzero rows; and $\operatorname{rank}(\cdot)$ denotes the rank of the matrix.

Unfortunately, solving model (2) is NP-hard. It was suggested in [46] to use the nuclear norm to relax the rank function and use the $l_{2,1}$ norm to relax the $l_{2,0}$ norm. The convex approximation of model (2) can be rewritten as follows:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \lambda \|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \end{aligned} \tag{3}$$

where $\|\mathbf{S}\|_* := \sum_{i=1}^{\operatorname{rank}(\mathbf{S})} \sigma_i(\mathbf{S})$ with $\sigma_i(\mathbf{S})$ being the ith singular value of $\mathbf{S}$; and $\|\mathbf{E}\|_{2,1} := \sum_{i=1}^{n} (\sum_{j=1}^{L} \mathbf{E}_{ij})^{1/2}$ with $\mathbf{E}_{ij}$ being the entry of $\mathbf{E}$.

C. Low-Rank and SR Model

To better characterize the local structure of each pixel, the sparse constraint has been introduced in the LRR formulation. The LRASR model can be expressed as follows [50]:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \beta \|\mathbf{S}\|_1 + \lambda \|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \end{aligned} \tag{4}$$

where $\beta, \lambda > 0$ denotes the regularization parameters; and $\|\cdot\|_1$ denotes the l_1 norm, i.e., the sum of the absolute value of all the entries.

D. Graph and TV-Regularized LRR Model

To preserve the spatial relationships and local geometrical structure in hyperspectral images, the graph and TV regularizers have been embedded in the LRR formulation. The GTVLRR model can be written as follows [57]:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} \|\mathbf{S}\|_* + \beta \operatorname{TV}(\mathbf{S}) + \gamma \operatorname{Tr}(\mathbf{SLS}^T) + \lambda \|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{DS} + \mathbf{E} \end{aligned} \tag{5}$$

where $\beta, \gamma, \lambda > 0$ denote the regularization parameters; and $\operatorname{TV}(\cdot)$ denotes the TV regularizer and is defined by the following equation:

$$\operatorname{TV}(\mathbf{S}) := \sum_{\{i,j\}\in\varepsilon} \|\mathbf{s}_i - \mathbf{s}_j\|_1 \tag{6}$$

where ε denotes the set of horizontal and vertical neighbors of $\mathbf{S}$; $\operatorname{Tr}(\mathbf{SLS}^T)$ denotes the graph regularizer, where $\mathbf{L} = \mathbf{D} - \mathbf{W}$ denotes the graph Laplacian matrix; $\mathbf{W}$ denotes the weight matrix of the graph and is defined by the following equation:

$$\mathbf{W}_{ij} = \exp\left(-\frac{\|\mathbf{s}_i - \mathbf{s}_j\|_2^2}{\sigma}\right) \tag{7}$$

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where $\mathbf{s}_i$ is among the k-nearest neighbors of $\mathbf{s}_j$; σ is a scalar parameter; and $\mathbf{D}$ denotes a diagonal matrix with entries measuring the degree of each pixel, and each $\mathbf{D}_{ij} = \sum_{j=1}^{n} \mathbf{W}_{ij}$.

The GTVLRR model can be viewed as a convex approximation of the following unified nonconvex model:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} \operatorname{rank}(\mathbf{S}) + \beta\ l_0\mathrm{TV}(\mathbf{S}) + \gamma\ \operatorname{Tr}(\mathbf{S}\mathbf{L}\mathbf{S}^T) + \lambda\ \|\mathbf{E}\|_{2,0} \\ &\text{s.t. } \mathbf{X} = \mathbf{D}\mathbf{S} + \mathbf{E} \end{aligned} \tag{8}$$

where l_0TV denotes the l_0 gradient and is defined by the following equation:

$$l_0\mathrm{TV}(\mathbf{S}) := \sum_{\{i,j\}\in\varepsilon} \|\mathbf{s}_i - \mathbf{s}_j\|_0 \tag{9}$$

where $\|\cdot\|_0$ counts the number of nonzero elements.

III. Proposed Approach and Solving Algorithm

Before stepping into the proposed approach, we first introduce the GSMs as they are the core ingredient in the framework. Then we derive the new nonconvex HADGSM model to approximate model (8) that accurately characterizes the intrinsic features in hyperspectral images.

A. Generalized Shrinkage Mappings

It is well-known that nonconvex models have been shown to give better results than convex ones in hyperspectral image processing. Herein, we design a nonconvex model using the GSMs to approximate the LRR-based anomaly detection model. To take advantage of the existing efficient algorithms, we expect to consider penalty functions with explicit proximal mappings. The GSMs we consider here first appeared in [66] to reconstruct MRI from very little data, and then were used in compressed sensing recovery [65] and hyperspectral unmixing task [63].

Definition 1: For a given matrix $\mathbf{C} \in \mathbb{R}^{m\times n}$ and $\lambda > 0$, the GSMs $S_{a,\lambda}$ and $S^g_{a,\lambda}$ from $\mathbb{R}^{m\times n}$ to $\mathbb{R}^{m\times n}$ are defined by

$$S_{a,\lambda}(\mathbf{C}) := \operatorname{sign}(\mathbf{C}) \circ s_{a,\lambda}(|\mathbf{C}|) \tag{10}$$

$$S^g_{a,\lambda}(\mathbf{C})(i,:) := \frac{\mathbf{C}(i,:)}{\|\mathbf{C}(i,:)\|_2} s^g_{a,\lambda}(\mathbf{C}(i,:)) \tag{11}$$

where $a \le 1$, $s_{a,\lambda}(\mathbf{C}) := \max(\mathbf{C} - \lambda^{2-a}\mathbf{C}^{a-1}, 0)$, and $s^g_{a,\lambda}(\mathbf{C}(i,:)) := \max(\|\mathbf{C}(i,:)\|_2 - \lambda^{2-a}\|\mathbf{C}(i,:)\|_2^{a-1}, 0)$ $(i = 1, 2, \ldots, m)$ denote the shrinkage functions.

Note that the GSMs degenerate into the proximal mappings of l_1 and $l_{2,1}$ norms in the case of $a = 1$, which can be formulated as follows:

$$\begin{aligned} \mathrm{Prox}_{\lambda\|\cdot\|_{1,1}} &:= \operatorname*{argmin}_{\mathbf{X}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda\|\mathbf{X}\|_1 \\ &= S_{1,\lambda}(\mathbf{C}) \end{aligned} \tag{12}$$

$$\begin{aligned} \mathrm{Prox}_{\lambda\|\cdot\|_{2,1}} &:= \operatorname*{argmin}_{\mathbf{X}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda\|\mathbf{X}\|_{2,1} \\ &= S^g_{1,\lambda}(\mathbf{C}). \end{aligned} \tag{13}$$

The following proposition introduced in [65] and [67] links the relationship between the GSMs and the induced nonconvex penalty functions.

Proposition 1: Let $\lambda > 0$, $a \in \mathbb{R}$. Assume $s_{a,\lambda} : [0,\infty) \to \mathbb{R}$ is continuous and satisfies $c \le \lambda$, then $s_{a,\lambda}(c) = 0$ for some $\lambda \ge 0$, is strictly increasing on $[\lambda, \infty)$ and $s_{a,\lambda}(c) \le c$. Define $S_{a,\lambda}(\mathbf{C}) := \operatorname{sign}(\mathbf{C}) \circ s_{a,\lambda}(|\mathbf{C}|)$. Then $S_{a,\lambda}$ is the proximal mapping of the penalty function $G(\mathbf{X}) = \sum_{i,j} g(\mathbf{X}_{ij})$, where g is even, strictly increasing and continuous on $[0,\infty)$, differentiable except at 0 with $\partial g(0) = [-1, 1]$ if and only if $\lambda > 0$. If $c - s(c)$ is nonincreasing on $[\lambda, \infty)$, then g is concave on $[0,\infty)$ and G satisfies the triangle inequality. Moreover, $S^g_{a,\lambda}$ is the proximal mapping of the penalty function $g(\|\mathbf{X}\|_{2,1})$.

The GSMs meet all the hypotheses of the above proposition for all the parameter values. The proof of this proposition is given in [67], where g is constructed using the Legendre–Fenchel transform of the antiderivative of s. However, the induced penalty functions are unable to be written in a closed-form expression since the nature of the Legendre–Fenchel transforms. Actually, it is much more useful for most state-of-the-art algorithms to know the explicit proximal mappings rather than the closed-form expression of the induced penalty functions. Note that the firm thresholding mapping and GSM were used in the unmixing tasks in [63]. However, the shrinkage mappings used in [63] only approximate the l_0 and low-rank regularizers, whereas the developed in this work is an extension version that can further approximate the l_0TV regularizer.

B. Hyperspectral Anomaly Detection via GSMs

The proposed HADGSM model uses the GSM-induced penalty functions to approximate model (8), which can be expressed as follows:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} G(\sigma(\mathbf{S})) + \beta\ G(\mathrm{TV}(\mathbf{S})) + \gamma\ \operatorname{Tr}(\mathbf{S}\mathbf{L}\mathbf{S}^T) + \lambda\ g(\|\mathbf{E}\|_{2,1}) \\ &\text{s.t. } \mathbf{X} = \mathbf{D}\mathbf{S} + \mathbf{E} \end{aligned} \tag{14}$$

where $\sigma(\mathbf{S})$ denotes the vector of singular values of $\mathbf{S}$. The goal of the HADGSM model is to jointly estimate the representation coefficients of the background $\mathbf{S}$ and the anomaly component $\mathbf{E}$, while the observed hyperspectral images $\mathbf{X}$ and the background dictionary $\mathbf{D}$ are given. Similar to the GTVLRR model, the proposed model uses the strategy proposed in [50] to construct the dictionary. In this strategy, K-means using the Euclidean distance is adopted to divide the hyperspectral images into M clusters first, followed by applying RX to each cluster. The square of the Mahalanobis distance between each test pixel and the local background mean in the cluster is calculated. Then P pixels with the smallest Mahalanobis distance are selected as the atoms in the dictionary.

The HADGSM model is capable of capturing the spectral correlation, spatial smoothness, and intrinsic local geometrical structure in hyperspectral images. More specifically, the first term of the objective function is the GSM-induced nuclear penalty, which is to express the spectral correlation sufficiently. While the second term of the objective function is the GSM-induced TV penalty, which is to further enhance the piecewise smoothness and maintain the sharp edges between the background and anomalies. The same graph penalty is also used in the third term to preserve the intrinsic manifold

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structure in hyperspectral images. The fourth term, on the other hand, is the GSM-induced group sparse penalty, which is to exact the nonzero columns of outliers more precisely. The constraint is the modeling term to enforce the reconstruction error. As a consequence, the proposed HADGSM model could be more accurate in representing the background part and anomaly part, providing a more accurate characterization of hyperspectral images. It is clear that the HADGSM model contains the GTVLRR model as a special case. Furthermore, the HADGSM model degenerates into the TV-regularized LRR (TVLRR) model [57] in the case of $\gamma = 0$ and $G = I$, and degenerates into the LRR model in the case of $\beta, \gamma = 0$ and $G = I$.

At first glance, it is quite challenging to solve the HADGSM model due to several nonconvex penalties involved in the optimization problem. Fortunately, the closed-form solutions of the nonconvex penalties in the model (14) can be given by GSMs, enabling most state-of-the-art algorithms such as ADMM to solve it. The ADMM which decomposes a complex problem into a series of more easily solvable subproblems has been extensively investigated in hyperspectral image processing [68], [69]. Before designing ADMM to solve the HADGSM model, we give a more detailed characterization of the TV norm. Define two linear operators $\mathcal{H}_h : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$ and $\mathcal{H}_v : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$ to compute the horizontal and vertical differences between the neighboring pixels of $\mathbf{S}$ as follows:

$$\mathcal{H}_h\mathbf{S} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_n], \quad \mathcal{H}_v\mathbf{S} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n]$$

where $\mathbf{c}_i = \mathbf{s}_i - \mathbf{s}_{i_h}$ and $\mathbf{b}_i = \mathbf{s}_i - \mathbf{s}_{i_v}$ $(i = 1, \ldots, n)$, with i denoting an index of a pixel, with i_h and i_v denoting the indices of the corresponding horizontal and vertical neighbors, respectively. Then the TV norm can be rewritten as follows:

$$\mathrm{TV}(\mathbf{S}) = \|\mathcal{H}\mathbf{S}\|_1 \tag{15}$$

where

$$\mathcal{H}\mathbf{S} = \begin{bmatrix} \mathcal{H}_h\mathbf{S} \\ \mathcal{H}_v\mathbf{S} \end{bmatrix}.$$

The problem (14) converts to the following equivalent form:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E}} G(\sigma(\mathbf{S})) + \beta\ G(\|\mathcal{H}\mathbf{S}\|_1) + \gamma\ \mathrm{Tr}(\mathbf{S}\mathbf{L}\mathbf{S}^T) + \lambda\ g(\|\mathbf{E}\|_{2,1}) \\ &\text{s.t. } \mathbf{X} = \mathbf{D}\mathbf{S} + \mathbf{E}. \end{aligned} \tag{16}$$

By introducing four slack variables $\mathbf{V}_1 \in \mathbb{R}^{m\times n}$, $\mathbf{V}_2 \in \mathbb{R}^{m\times n}$, $\mathbf{V}_3 \in \mathbb{R}^{2m\times n}$, and $\mathbf{V}_4 \in \mathbb{R}^{m\times n}$, the problem (16) converts to the following equivalent problem:

$$\begin{aligned} &\min_{\mathbf{S},\mathbf{E},\mathbf{V}_i} G(\sigma(\mathbf{V}_1)) + \beta\ G(\|\mathbf{V}_3\|_1) + \gamma\ \mathrm{Tr}\big(\mathbf{V}_4\mathbf{L}\mathbf{V}_4^T\big) \\ &\qquad + \lambda\ g(\|\mathbf{E}\|_{2,1}) \\ &\text{s.t. } \mathbf{X} = \mathbf{D}\mathbf{S} + \mathbf{E}, \quad \mathbf{V}_1 = \mathbf{S},\ \mathbf{V}_2 = \mathbf{S} \\ &\qquad \mathbf{V}_3 = \mathcal{H}\mathbf{V}_2,\ \mathbf{V}_4 = \mathbf{S}, \quad i = 1, \ldots, 4. \end{aligned} \tag{17}$$

The above problem can also be rewritten in the following form:

$$\begin{aligned} &\min_{\mathbf{M},\mathbf{N}} f_1(\mathbf{M}) + f_2(\mathbf{N}) \\ &\text{s.t. } \mathcal{A}\mathbf{M} + \mathcal{B}\mathbf{N} = \mathcal{C} \end{aligned} \tag{18}$$

where

$$\begin{aligned} \mathbf{M} &= \big(\mathbf{S}^T, \mathbf{V}_3^T\big)^T \in \mathbb{R}^{3m\times n} \\ \mathbf{N} &= \big(\mathbf{V}_1^T, \mathbf{V}_2^T, \mathbf{V}_4^T, \mathbf{E}^T\big)^T \in \mathbb{R}^{(3m+L)\times n} \\ f_1(\mathbf{M}) &= \beta\ G(\|\mathbf{V}_3\|_1) \\ f_2(\mathbf{N}) &= G(\sigma(\mathbf{V}_1)) + \gamma\ \mathrm{Tr}\big(\mathbf{V}_4\mathbf{L}\mathbf{V}_4^T\big) + \lambda\ g(\|\mathbf{E}\|_{2,1}) \end{aligned}$$

and

$$\mathcal{A} = \begin{pmatrix} \mathbf{D}_{L\times m} & 0_{L\times 2m} \\ \mathbf{I}_{m\times m} & 0_{m\times 2m} \\ \mathbf{I}_{m\times m} & 0_{m\times 2m} \\ 0_{2m\times m} & \mathbf{I}_{2m\times 2m} \\ \mathbf{I}_{m\times m} & 0_{m\times 2m} \end{pmatrix}$$

$$\mathcal{B} = \begin{pmatrix} 0_{L\times m} & 0_{L\times m} & 0_{L\times m} & \mathbf{I}_{L\times L} \\ -\mathbf{I}_{m\times m} & 0_{m\times m} & 0_{m\times m} & 0_{m\times L} \\ 0_{m\times m} & -\mathbf{I}_{m\times m} & 0_{m\times m} & 0_{m\times L} \\ 0_{2m\times m} & -\mathcal{H} & 0_{2m\times m} & 0_{2m\times L} \\ 0_{m\times m} & 0_{m\times m} & -\mathbf{I}_{m\times m} & 0_{m\times L} \end{pmatrix}$$

$$\mathcal{C} = (\mathbf{X}^T, 0_{n\times m}, 0_{n\times m}, 0_{n\times 2m}, 0_{n\times m})^T.$$

Let $\mu > 0$ be a penalty parameter, the augmented Lagrangian function of (18) can be written as follows:

$$\begin{aligned} L_\mu(\mathbf{M}, \mathbf{N}; \mathbf{U}) &= f_1(\mathbf{M}) + f_2(\mathbf{N}) \\ &\quad + \frac{\mu}{2}\|\mathcal{A}\mathbf{M} + \mathcal{B}\mathbf{N} - \mathcal{C} - \mu^{-1}\mathbf{U}\|_F^2 - \frac{1}{2\mu}\|\mathbf{U}\|_F^2 \end{aligned} \tag{19}$$

where $\mathbf{U} = (\mathbf{U}_1^T, \mathbf{U}_2^T, \mathbf{U}_3^T, \mathbf{U}_4^T, \mathbf{U}_5^T)^T$ are the Lagrangian dual variables.

From (18), we can observe that the optimization problem contains two blocks $\mathbf{M}$ and $\mathbf{N}$, and the variables in each block are separable and independent. Thereby, the classical ADMM framework can be used by updating one block while fixing the others. The iterations go as follows:

$$\mathbf{M}^{k+1} = \operatorname*{argmin}_{\mathbf{M}} L_\mu(\mathbf{M}, \mathbf{N}^k; \mathbf{U}^k) \tag{20}$$

$$\mathbf{N}^{k+1} = \operatorname*{argmin}_{\mathbf{N}} L_\mu(\mathbf{M}^{k+1}, \mathbf{N}; \mathbf{U}^k) \tag{21}$$

$$\mathbf{U}^{k+1} = \mathbf{U}^k + \mu(\mathcal{A}\mathbf{M}^{k+1} + \mathcal{B}\mathbf{N}^{k+1} - \mathcal{C}). \tag{22}$$

The details are addressed as follows. The expanded form of (19) can be written as follows:

$$\begin{aligned} &L_\mu(\mathbf{S}, \mathbf{E}, \mathbf{V}_i; \mathbf{U}_j) \\ &\quad = G(\sigma(\mathbf{V}_1)) + \beta\ G(\|\mathbf{V}_3\|_1) + \gamma\ \mathrm{Tr}\big(\mathbf{V}_4\mathbf{L}\mathbf{V}_4^T\big) \\ &\qquad + \lambda g(\|\mathbf{E}\|_{2,1}) + \frac{\mu}{2}\|\mathbf{X} - \mathbf{D}\mathbf{S} - \mathbf{E} \\ &\qquad - \mu^{-1}\mathbf{U}_1\|_F^2 + \frac{\mu}{2}\|\mathbf{V}_1 - \mathbf{S} - \mu^{-1}\mathbf{U}_2\|_F^2 \\ &\qquad + \frac{\mu}{2}\|\mathbf{V}_2 - \mathbf{S} - \mu^{-1}\mathbf{U}_3\|_F^2 + \frac{\mu}{2}\|\mathbf{V}_3 \\ &\qquad - \mathcal{H}\mathbf{V}_2 - \mu^{-1}\mathbf{U}_4\|_F^2 + \frac{\mu}{2}\|\mathbf{V}_4 - \mathbf{S} \\ &\qquad - \mu^{-1}\mathbf{U}_5\|_F^2 - \frac{1}{2\mu}\|\mathbf{U}_1\|_F^2 - \frac{1}{2\mu}\|\mathbf{U}_2\|_F^2 \\ &\qquad - \frac{1}{2\mu}\|\mathbf{U}_3\|_F^2 - \frac{1}{2\mu}\|\mathbf{U}_4\|_F^2 - \frac{1}{2\mu}\|\mathbf{U}_5\|_F^2 \\ &\qquad\qquad i = 1, \ldots, 4, \quad j = 1, \ldots, 5 \end{aligned} \tag{23}$$

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where $\mathbf{U}_1 \in \mathbb{R}^{L\times n}$, $\mathbf{U}_2 \in \mathbb{R}^{m\times n}$, $\mathbf{U}_3 \in \mathbb{R}^{m\times n}$, $\mathbf{U}_4 \in \mathbb{R}^{2m\times n}$, and $\mathbf{U}_5 \in \mathbb{R}^{m\times n}$.

1) Update $\boldsymbol{S}^{k+1}$: The minimization of (23) with respect to $\mathbf{S}$ can be formulated as follows:

$$\begin{aligned}\mathbf{S}^{k+1} &= \underset{\mathbf{S}}{\operatorname{argmin}} L_\mu\left(\mathbf{S}, \mathbf{E}^k, \mathbf{V}_i^k; \mathbf{U}_j^k\right)\\ &= \underset{\mathbf{S}}{\operatorname{argmin}}\Big\{\frac{\mu}{2}\|\mathbf{X}-\mathbf{D}\mathbf{S}-\mathbf{E}^k-\mu^{-1}\mathbf{U}_1^k\|_F^2\\ &\quad+\frac{\mu}{2}\|\mathbf{V}_1^k-\mathbf{S}-\mu^{-1}\mathbf{U}_2^k\|_F^2+\frac{\mu}{2}\|\mathbf{V}_2^k\\ &\quad-\mathbf{S}-\mu^{-1}\mathbf{U}_3^k\|_F^2+\frac{\mu}{2}\left\|\mathbf{V}_4^k-\mathbf{S}-\mu^{-1}\mathbf{U}_5^k\right\|_F^2\Big\}.\end{aligned} \tag{24}$$

By simple calculation, the closed-form solution of (24) can be obtained by the following equation:

$$\begin{aligned}\mathbf{S}^{k+1} = (\mathbf{D}^T\mathbf{D}+3\mathbf{I})^{-1}\big[\mathbf{D}^T(\mathbf{X}-\mu^{-1}\mathbf{U}_1^k-\mathbf{E}^k)+\mathbf{V}_1^k\\ -\mu^{-1}\mathbf{U}_2^k+\mathbf{V}_2^k-\mu^{-1}\mathbf{U}_3^k+\mathbf{V}_4^k-\mu^{-1}\mathbf{U}_5^k\big].\end{aligned} \tag{25}$$

2) Update $\boldsymbol{V}_3^{k+1}$: The minimization of (23) with respect to $\mathbf{V}_3$ can be formulated as follows:

$$\begin{aligned}\mathbf{V}_3^{k+1} &= \underset{\mathbf{V}_3}{\operatorname{argmin}} L_\mu\left(\mathbf{S}^{k+1}, \mathbf{E}^k, \mathbf{V}_1^k, \mathbf{V}_2^k, \mathbf{V}_3, \mathbf{V}_4^k; \mathbf{U}_j^k\right)\\ &= \underset{\mathbf{V}_3}{\operatorname{argmin}}\Big\{\beta\ G(\|\mathbf{V}_3\|_1)+\frac{\mu}{2}\|\mathbf{V}_3-\mathcal{H}\mathbf{V}_2^k-\mu^{-1}\mathbf{U}_4^k\|_F^2\Big\}.\end{aligned} \tag{26}$$

By Proposition 1, the closed-form solution of (26) can be derived as follows:

$$\mathbf{V}_3^{k+1} = S_{a,\frac{\beta}{\mu}}\left(\mathcal{H}\mathbf{V}_2^k+\mu^{-1}\mathbf{U}_4^k\right). \tag{27}$$

3) Update $\mathcal{V}_1^{k+1}$: The minimization of (23) to $\mathbf{V}_1$ can be written as

$$\begin{aligned}\mathbf{V}_1^{k+1} &= \underset{\mathbf{V}_1}{\operatorname{argmin}} L_\mu\left(\mathbf{S}^{k+1}, \mathbf{E}^k, \mathbf{V}_1, \mathbf{V}_2^k, \mathbf{V}_3^{k+1}, \mathbf{V}_4^k; \mathbf{U}_j^k\right)\\ &= \underset{\mathbf{V}_1}{\operatorname{argmin}}\Big\{G(\sigma(\mathbf{V}_1))+\frac{\mu}{2}\|\mathbf{V}_1-\mathbf{S}^{k+1}-\mu^{-1}\mathbf{U}_2^k\|_F^2\Big\}.\end{aligned} \tag{28}$$

The following proposition proved in [70] turns out (28) which can be solved efficiently by means of the GSM:

Proposition 2: Let G be defined by Proposition 1 and the matrix $\mathbf{C}$ has singular value decomposition (SVD): $\mathbf{C} = \mathbf{P}\Sigma\mathbf{Q}^T$. Then the solution to the following optimization problem

$$\underset{\mathbf{V}_1}{\operatorname{argmin}}\left\{\mu G(\sigma(\mathbf{V}_1))+\frac{1}{2}\|\mathbf{V}_1-\mathbf{C}\|_F^2\right\} \tag{29}$$

can be given by $\mathbf{V}_1 = \mathbf{P}S_{a,\mu}(\Sigma)\mathbf{Q}^T$.

Then the solution of (28) can be given by Proposition 2 in a closed form as follows:

$$\mathbf{V}_1^{k+1} = \mathbf{P}S_{a,\frac{1}{\mu}}(\Sigma)\mathbf{Q}^T \tag{30}$$

where $\mathbf{P}\Sigma\mathbf{Q}^T = \mathbf{S}^{k+1}+\mu^{-1}\mathbf{U}_2^k$.

4) Update $\boldsymbol{V}_2^{k+1}$: The minimization of (23) to $\mathbf{V}_2$ can be formulated as follows:

$$\begin{aligned}\mathbf{V}_2^{k+1} &= \underset{\mathbf{V}_2}{\operatorname{argmin}} L_\mu\left(\mathbf{S}^{k+1}, \mathbf{E}^k, \mathbf{V}_1^{k+1}, \mathbf{V}_2, \mathbf{V}_3^{k+1}, \mathbf{V}_4^k; \mathbf{U}_j^k\right)\\ &= \underset{\mathbf{V}_2}{\operatorname{argmin}}\Big\{\frac{\mu}{2}\|\mathbf{V}_2-\mathbf{S}^{k+1}-\mu^{-1}\mathbf{U}_3^k\|_F^2\\ &\quad+\frac{\mu}{2}\|\mathbf{V}_3^{k+1}-\mathcal{H}\mathbf{V}_2-\mu^{-1}\mathbf{U}_4^k\|_F^2\Big\}.\end{aligned} \tag{31}$$

Then the solution of (31) can be given by the following equation:

$$\begin{aligned}\mathbf{V}_2^{k+1} = (\mathcal{H}^T\mathcal{H}+\mathbf{I})^{-1}\\ \times\left[\mathbf{S}^{k+1}+\mu^{-1}\mathbf{U}_3^k+\mathcal{H}^T\left(\mathbf{V}_3^{k+1}-\mu^{-1}\mathbf{U}_4^k\right)\right]\end{aligned} \tag{32}$$

which can be computed via the discrete Fourier transform diagonalization.

5) Update $\boldsymbol{V}_4^{k+1}$: The minimization of (23) to $\mathbf{V}_4$ can be formulated as follows:

$$\begin{aligned}\mathbf{V}_4^{k+1} &= \underset{\mathbf{V}_4}{\operatorname{argmin}} L_\mu\left(\mathbf{S}^{k+1}, \mathbf{E}^k, \mathbf{V}_1^{k+1}, \mathbf{V}_2^{k+1}, \mathbf{V}_3^{k+1}, \mathbf{V}_4; \mathbf{U}_j^k\right)\\ &= \underset{\mathbf{V}_4}{\operatorname{argmin}}\Big\{\gamma\ \operatorname{Tr}(\mathbf{V}_4\mathbf{L}\mathbf{V}_4^T)+\frac{\mu}{2}\left\|\mathbf{V}_4-\mathbf{S}^{k+1}-\mu^{-1}\mathbf{U}_5^k\right\|_F^2\Big\}.\end{aligned} \tag{33}$$

The solution of (33) can be obtained by solving the following linear system of equations:

$$\mathbf{V}_4^{k+1}(2\gamma\mathbf{L}+\mu\mathbf{I}) = \mu\mathbf{S}^{k+1}+\mathbf{U}_5^k. \tag{34}$$

Equation (34) is a large-scale sparse system of equations that can be efficiently solved by preconditioned conjugate gradient.

6) Update $\boldsymbol{E}^{k+1}$: The minimization of (23) with respect to $\mathbf{E}$ is performed as follows:

$$\begin{aligned}\mathbf{E}^{k+1} &= \underset{\mathbf{E}}{\operatorname{argmin}} L_\mu\left(\mathbf{S}^{k+1}, \mathbf{E}, \mathbf{V}_i^{k+1}; \mathbf{U}_j^k\right)\\ &= \underset{\mathbf{E}}{\operatorname{argmin}}\ \lambda\ g(\|\mathbf{E}\|_{2,1})+\frac{\mu}{2}\left\|\mathbf{X}-\mathbf{D}\mathbf{S}^{k+1}-\mathbf{E}-\mu^{-1}\mathbf{U}_1^k\right\|_F^2.\end{aligned} \tag{35}$$

The solution of (35) can be given by Proposition 1 in a closed form as follows:

$$\mathbf{E}^{k+1} = S^g_{a,\frac{\lambda}{\mu}}\left(\mathbf{X}-\mathbf{D}\mathbf{S}^{k+1}-\mu^{-1}\mathbf{U}_1^k\right). \tag{36}$$

7) Update $\boldsymbol{U}_1^{k+1}, \boldsymbol{U}_2^{k+1}, \boldsymbol{U}_3^{k+1}, \boldsymbol{U}_4^{k+1}, \boldsymbol{U}_5^{k+1}$: The Lagrangian dual variables are updated as follows:

$$\begin{aligned}\mathbf{U}_1^{k+1} &= \mathbf{U}_1^k-\mu\left(\mathbf{X}-\mathbf{D}\mathbf{S}^{k+1}-\mathbf{E}^{k+1}\right)\\ \mathbf{U}_2^{k+1} &= \mathbf{U}_2^k-\mu\left(\mathbf{V}_1^{k+1}-\mathbf{S}^{k+1}\right)\\ \mathbf{U}_3^{k+1} &= \mathbf{U}_3^k-\mu\left(\mathbf{V}_2^{k+1}-\mathbf{S}^{k+1}\right)\\ \mathbf{U}_4^{k+1} &= \mathbf{U}_4^k-\mu\left(\mathbf{V}_3^{k+1}-\mathcal{H}\mathbf{V}_2^{k+1}\right)\\ \mathbf{U}_5^{k+1} &= \mathbf{U}_5^k-\mu\left(\mathbf{V}_4^{k+1}-\mathbf{S}^{k+1}\right).\end{aligned} \tag{37}$$

Once the optimization for (14) is finished, the detection task is performed on the anomaly component. More specifically, the anomaly map $\mathbf{T}$ can be determined by the response of $\mathbf{E}$ as follows:

$$\mathbf{T}(\mathbf{x}_i) = \|\mathbf{E}(:,i)\|_2 > \epsilon. \tag{38}$$

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Algorithm 1 HADGSM

Input: Hyperspectral image $\mathbf{X}$, dictionary $\mathbf{D}$, and the regularization parameters β, γ, λ.

1: **Initialization**: $\mathbf{M}^0 = 0$, $\mathbf{N}^0 = 0$, $\mathbf{U}^0 = 0$, $k = 0$, $\mu^0 = 10^{-4}$, $\mu_{max} = 10^{10}$, $\rho = 1.5$, maxiter = 50;
2: **while** $0 \leq k \leq$ maxiter or meet the convergence conditions,
3: Update $\mathbf{M}^{k+1} = (\mathbf{S}^{k+1}, \mathbf{V}_3^{k+1})$ by (25) and (27);
4: Update $\mathbf{N}^{k+1} = (\mathbf{V}_1^{k+1}, \mathbf{V}_2^{k+1}, \mathbf{V}_4^{k+1}, \mathbf{E}^{k+1})$ by (30), (32), (34), and (36);
5: Update $\mathbf{U}^{k+1} = (\mathbf{U}_1^{k+1}, \mathbf{U}_2^{k+1}, \mathbf{U}_3^{k+1}, \mathbf{U}_4^{k+1}, \mathbf{U}_5^{k+1})$ by (37);
6: Update $\mu^{k+1} = \min(\rho\mu^k, \mu_{max})$;
7: $k = k + 1$.
8: **end while**
9: Obtain the optimal solution $\mathbf{E}^*$;
10: Compute the anomaly detection map $\mathbf{T}$ by (38);

Output: Anomaly detection map $\mathbf{T}$.

The ith pixel is claimed to be an anomaly pixel if (38) is larger than a predefined threshold ϵ. The algorithm is terminated if the maximum number of iterations is achieved or the stopping criterion

$$\|\mathcal{A}\mathbf{M} + \mathcal{B}\mathbf{N} - \mathcal{C}\|_F \leq \eta \tag{39}$$

is satisfied, where η is set to $\sqrt{n} \times 10^{-5}$. The whole procedure of the proposed framework is summarized in Algorithm 1.

IV. Numerical Experiments

To evaluate the proposed framework, several real datasets with ground-truth maps have been used in this section. The results of some classic methods such as RX [27], RPCA-RX [48], TPCA [58], LRR [45], [46], LRASR [50], TVLRR [57], and GTVLRR [57] were exhibited for comparison purposes. The number of nearest neighbors k was set to 10, and the scale parameter σ was set to 1 for the graph regularizer in the GTVLRR and HADGSM models. To improve the efficiency of the detection, the principal component analysis (PCA) method was applied as one preprocessing step for all the approaches. For a fair comparison, the parameters of all the compared methods were set as the recommended values or turned to obtain the best performance. In the dictionary construction step, M and P are set to 15 and 20, respectively, and the number of atoms is 300. All the experiments were implemented in MATLAB R2017b on a computer with 2.1-GHz Inter Core i7 and 64-GB memory. To quantitatively assess the detection accuracy of the different methods, the receiver operating characteristic (ROC) curve was used as the objective metric. The ROC curve plots the probability of detection as a function of the false alarm rate under different segmentation thresholds. In general, the closer the ROC curve is to the upper leftmost corner, the better the detection performance. In addition, the area under the ROC curve (AUC) was also calculated to quantitatively compare the different methods. The larger AUC values indicate the better performance of detection.

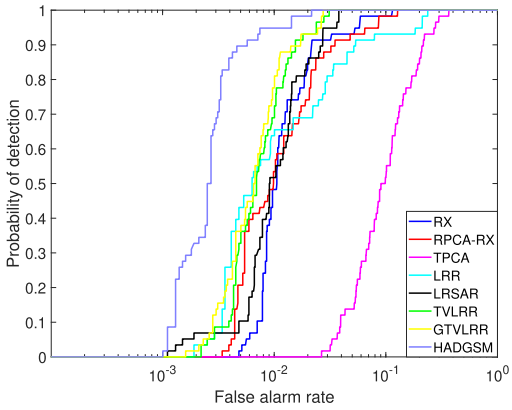


Fig. 1. ROC curves obtained by different methods for the San Diego dataset.

A. Experiments on San Diego Dataset

1) Data Description: The first hyperspectral data we used were the San Diego dataset, which is quite prevalent for hyperspectral detection tasks. We selected a 100 × 100 pixels data subset at the left-top region to evaluate the detection performance between the proposed method and the compared approaches. This dataset was captured by the airborne visible/infrared imaging spectrometer (ARIVIS) sensor over the area located in San Diego, USA. The three airplanes are regarded as anomalies, and the roofs, grass, buildings, and shadows are regarded as the background. The water vapor absorption and noise bands were eliminated, leaving a total of 189 bands in this scene. The false-color image and ground-truth map of the San Diego dataset are shown in Fig. 2(a) and (b), respectively.

2) Experimental Setup: The PCA method is used as the preprocessing step for all the approaches to improve the efficiency of the detection tasks. After careful tuning, most approaches achieve the best detection performance when the dimension of the San Diego dataset is set to 6. To fairly assess the detection performance, the optimal parameters for all the LRR-based approaches are set as follows. The group sparsity regularization of the LRR model is parameterized by 0.05. The sparsity and group sparsity regularizations for the LRASR model are parameterized by {1, 1}. The parameters for the TVLRR model are set as {0.2, 0.5}, and for the GTVLRR model are set as {0.2, 0.05, 0.5}. The parameters of the proposed HADGSM model are set as {2, 0.5, 2.5}.

3) Results and Analysis: Fig. 1 shows plots of the ROC curves obtained by different methods for the San Diego dataset. From Fig. 1, one can observe that the TPCA method yields the lowest probability of detection, while the RX, RPCA-RX, LRR, and LRSAR methods generate better detection performance. The TVLRR and GTVLRR methods further improve the probability of detection by exploiting the spatial information. The proposed HADGSM method achieves the highest probability of detection for all the false alarm rate values. The AUC values of different methods are provided in Table I. The proposed HADGSM model achieves the largest AUC score as expected. The computation costs of all the methods are also compared in Table I. The RX and RPCA-RX methods are fast due to their simple implementation. The

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TABLE I
AUC VALUES AND RUNNING TIME OF DIFFERENT METHODS FOR THE SAN DIEGO DATASET

Metric	Ideal Value	Method							
		RX [27]	RPCA-RX [48]	TPCA [58]	LRR [46]	LRASR [50]	TVLRR [57]	GTVLRR [57]	HADGSM
AUC	1	0.9844	0.9826	0.8849	0.9726	0.9865	0.9913	0.9920	**0.9967**
Time (Sec.)	-	**0.01**	0.11	9.66	24.72	48.13	81.23	85.54	38.85

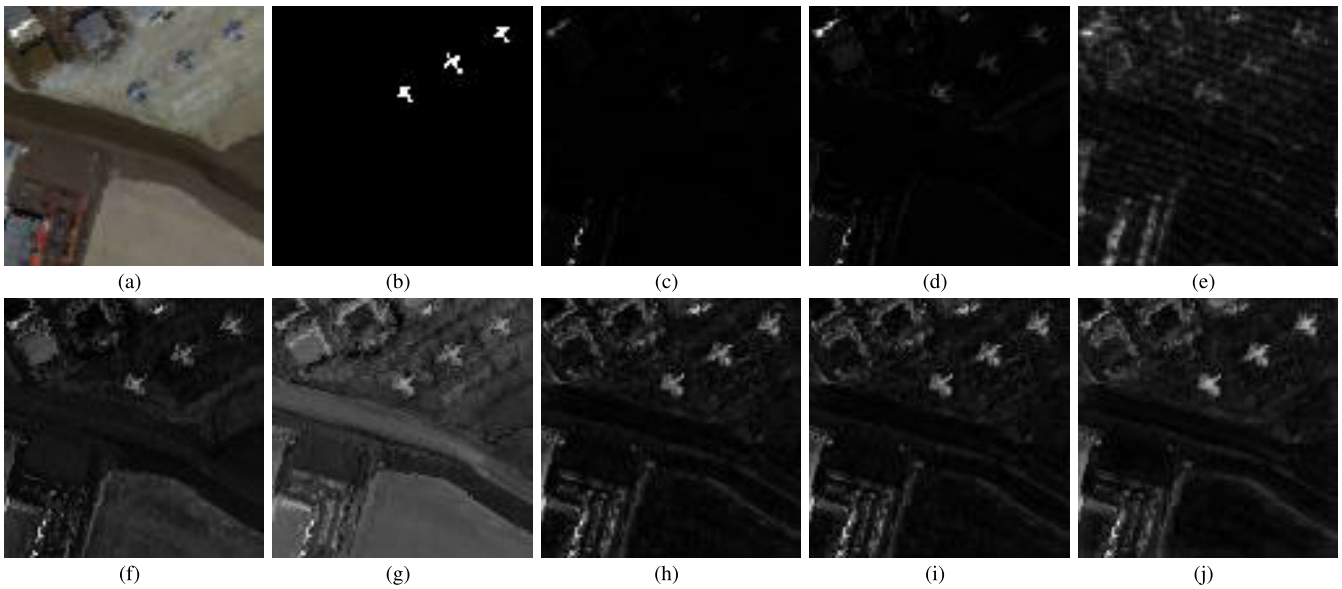


Fig. 2. Detection maps of different methods for the San Diego dataset. (a) False-color image. (b) Ground-truth map. (c) RX. (d) RPCA-RX. (e) TPCA. (f) LRR. (g) LRASR. (h) TVLRR. (i) GTVLRR. (j) HADGSM.

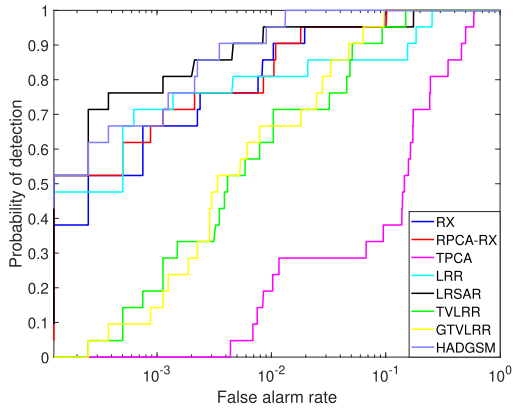


Fig. 3. ROC curves obtained by different methods for the Hydice-Urban dataset.

LRR-based methods cost more time since hundreds of iterations are required to converge. Among them, the GTVLRR method is the most time-consuming, while our method only takes less than half the time of the GTVLRR method.

A visual representation of the detection maps obtained by different methods for the San Diego dataset is shown in Fig. 2. One can clearly observe that the RX method failed to find the anomalies, and the RPCA-RX and TPCA methods detected anomalies but failed to detect the contours of the airplanes. All the LRR-based methods successfully detect the airplanes, among which the proposed HADGSM method provides a clearer shape of the airplanes than other methods. These observations confirm the advantage of the proposed method to not only the statistical-based approaches but also the LRR-based approaches.

B. Experiments on Hydice-Urban Dataset

1) Data Description: The second hyperspectral data used for detection were collected by the hyperspectral digital imagery collection experiment (HYDICE) sensor over an urban area in CA, USA. This Hydice-Urban dataset consists of 80 × 100 pixels and 175 bands, with a 1-m spatial resolution and 10-nm spectral resolution. Anomalies consist of vehicles and roofs of different sizes in this scene. The false-color image and ground-truth map of the Hydice-Urban dataset are shown in Fig. 4(a) and (b), respectively.

2) Experimental Setup: Similarly, the PCA method is adopted first to reduce the dimension of the Hydice-Urban dataset and set it as 17. We record the optimal parameters for the LRR-based methods as follows. For the LRR method, the group sparsity regularization is parameterized by 0.5. For the LRASR method, the sparsity and group sparsity regularizations are parameterized by {0.1, 1}. For the TVLRR method, the TV and group sparsity regularizations are parameterized

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TABLE II
AUC VALUES AND RUNNING TIME OF DIFFERENT METHODS FOR THE HYDICE-URBAN DATASET

Metric	Ideal Value	Method							
		RX [27]	RPCA-RX [48]	TPCA [58]	LRR [46]	LRASR [50]	TVLRR [57]	GTVLRR [57]	HADGSM
AUC	1	0.9927	0.9926	0.8242	0.9703	0.9908	0.9774	0.9833	**0.9984**
Time (Sec.)	-	**0.01**	0.14	6.69	19.93	26.68	74.23	66.54	31.23

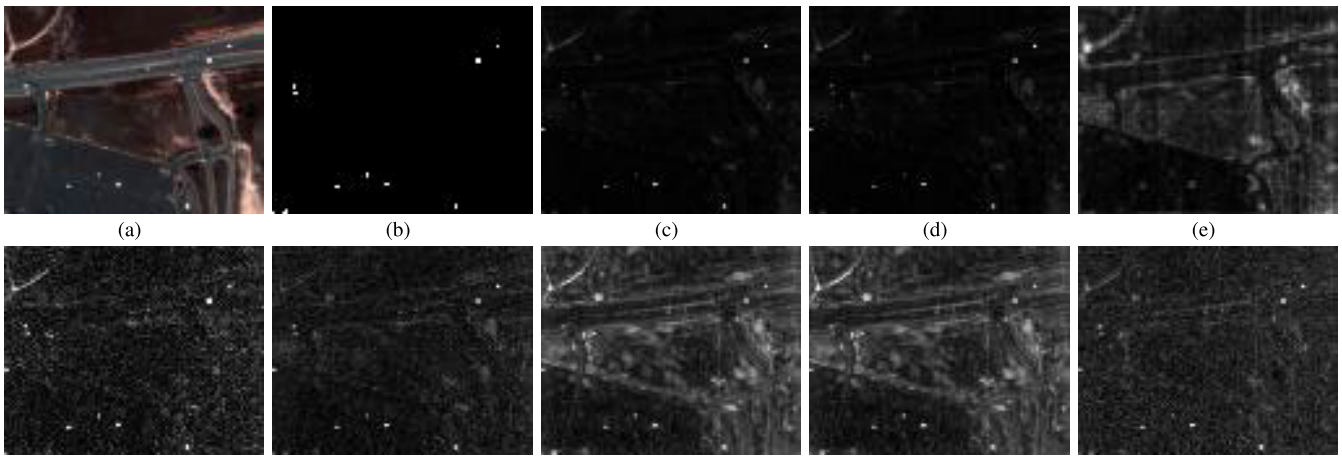


Fig. 4. Detection maps of different methods for the Hydice-Urban dataset. (a) False-color image. (b) Ground-truth map. (c) RX. (d) RPCA-RX. (e) TPCA. (f) LRR. (g) LRASR. (h) TVLRR. (i) GTVLRR. (j) HADGSM.

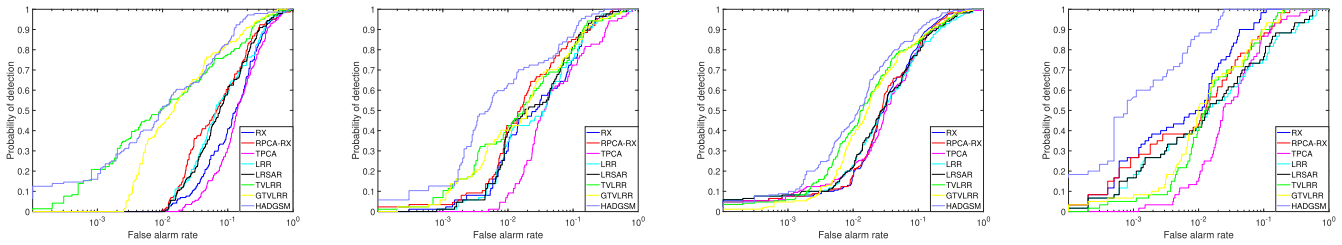


Fig. 5. ROC curves obtained by different methods for the Airport dataset. (a) Airport-1. (b) Airport-2. (c) Airport-3. (d) Airport-4.

by {0.2, 0.5}. For the GTVLRR method, the parameters for TV, group sparsity, and graph regularizations are set as {0.2, 0.05, 0.5}. The parameters of the proposed HADGSM method are set as {0.1, 0.5, 1.5}.

3) Results and Analysis: Fig. 3 shows plots of the ROC curves obtained by different methods for the Hydice-Urban dataset. From Fig. 3, we can observe that the LRASR and HADGSM methods achieve a higher probability of detection values than other methods. The probability of detection of the HADGSM method is smaller than the LRASR method in the beginning. When the false alarm rate is larger than 10^{-2}, the proposed HADGSM method achieves 100% detection. Table II lists the AUC values and running time of different methods for the Hydice-Urban dataset. The AUC value of the HADGSM method is still the highest among all the compared methods. Furthermore, the proposed method also exhibits decent competitiveness compared with other LRR-based methods in terms of running time.

For visual comparison, Fig. 4 shows the detection maps obtained by different methods for the Hydice-Urban dataset. The RX and RPCA-RX methods show neat detection maps but miss several targets. The contour of the bridge in the image is retained by the TPCA method, but most targets are ignored. Although the targets are detected by the TVLRR and GTVLRR methods, the false alarm rate is too high. The LRR, LRASR, and HADGSM methods show similar detection maps, among which the HADGSM method achieves better performance than the former methods. These observations further reveal the superiority of the proposed nonconvex framework for detection problems.

C. Experiments on Airport Dataset

1) Data Description: The third hyperspectral data we used were the Airport dataset captured by the ARIVIS sensor over the area located in Los Angeles and Gulfport, USA. These data contain four airport scenes, each with a size of

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TABLE III
AUC VALUES AND RUNNING TIME OF DIFFERENT METHODS FOR THE AIRPORT DATASET

Data	Metric	Ideal Value	Method							
			RX [27]	RPCA-RX [48]	TPCA [58]	LRR [46]	LRASR [50]	TVLRR [57]	GTVLRR [57]	HADGSM
Airport-1	AUC	1	0.8222	0.8717	0.8023	0.8584	0.8647	0.9260	0.9387	**0.9497**
	Time (Sec.)	-	**0.02**	0.11	10.86	11.83	16.82	80.62	89.32	42.91
Airport-2	AUC	1	0.9324	0.9486	0.8891	0.9289	0.9388	0.9389	0.9403	**0.9693**
	Time (Sec.)	-	**0.02**	0.10	11.07	11.36	16.81	79.78	79.21	42.71
Airport-3	AUC	1	0.9399	0.9374	0.9297	0.9212	0.9388	0.9453	0.9456	**0.9636**
	Time (Sec.)	-	**0.02**	0.19	11.02	7.84	16.81	69.28	67.58	41.31
Airport-4	AUC	1	0.9813	0.9638	0.9432	0.9142	0.9242	0.9645	0.9659	**0.9959**
	Time (Sec.)	-	**0.01**	0.18	10.19	11.62	14.28	74.11	73.98	39.97
	Average AUC	1	0.9189	0.9304	0.8911	0.9057	0.9166	0.9437	0.9476	**0.9696**

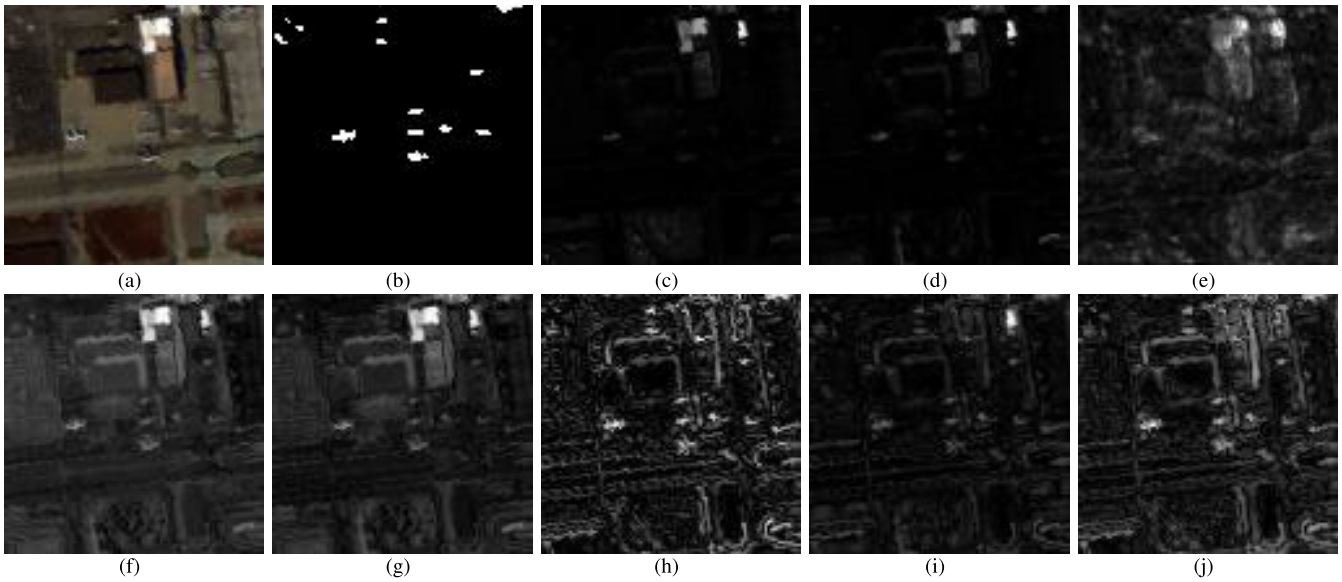


Fig. 6. Detection maps of different methods for the Airport-1 dataset. (a) False-color image. (b) Ground-truth map. (c) RX. (d) RPCA-RX. (e) TPCA. (f) LRR. (g) LRASR. (h) TVLRR. (i) GTVLRR. (j) HADGSM.

100 × 100 pixels, measured in 224 bands ranging from 400 to 2500 nm with a 10-nm spectral resolution. In those scenes, anomalies are flights of different heights, while buildings or roads are regarded as the background. Before the analysis, the atmospheric effects and dense water vapor absorption bands were removed, leaving a total of 205 bands. The false-color images and ground-truth maps of the Airport dataset are shown in Figs. 6–9(a) and (b), respectively.

2) Experimental Setup: Likewise, the PCA method was used as the preprocessing step. The dimensions of the four airport datasets are set to 4, 5, 17, and 17, respectively. The optimal parameters of all the LRR-based approaches for four airport datasets are recorded as follows. The parameters for Airport-1 are 0.001 for the LRR method, {0.001, 0.005} for the LRASR method, {0.2, 0.5} for the TVLRR method, {0.2, 0.05, 0.5} for the GTVLRR method, and {2.5, 0.1, 3} for the HADGSM method. The parameters for Airport-2 are 0.005 for the LRR method, {0.001, 0.001} for the LRASR method, {0.2, 0.5} for the TVLRR method, {0.2, 0.05, 0.5} for the GTVLRR method, and {2, 1, 1.5} for the HADGSM method. The parameters for Airport-3 are 0.001 for the LRR method, {0.001, 0.005} for the LRASR method, {0.2, 0.5} for the TVLRR method, {0.2, 0.05, 0.5} for the GTVLRR method, and {1, 0.1, 5.5} for the HADGSM method. The parameters for Airport-4 are 0.005 for the LRR method, {0.001, 0.1} for the LRASR method, {0.2, 0.5} for the TVLRR method, {0.2, 0.05, 0.5} for the GTVLRR method, and {0.5, 1, 5.5} for the HADGSM method.

3) Results and Analysis: The ROC curves obtained by different methods for the Airport dataset are plotted in Fig. 5.

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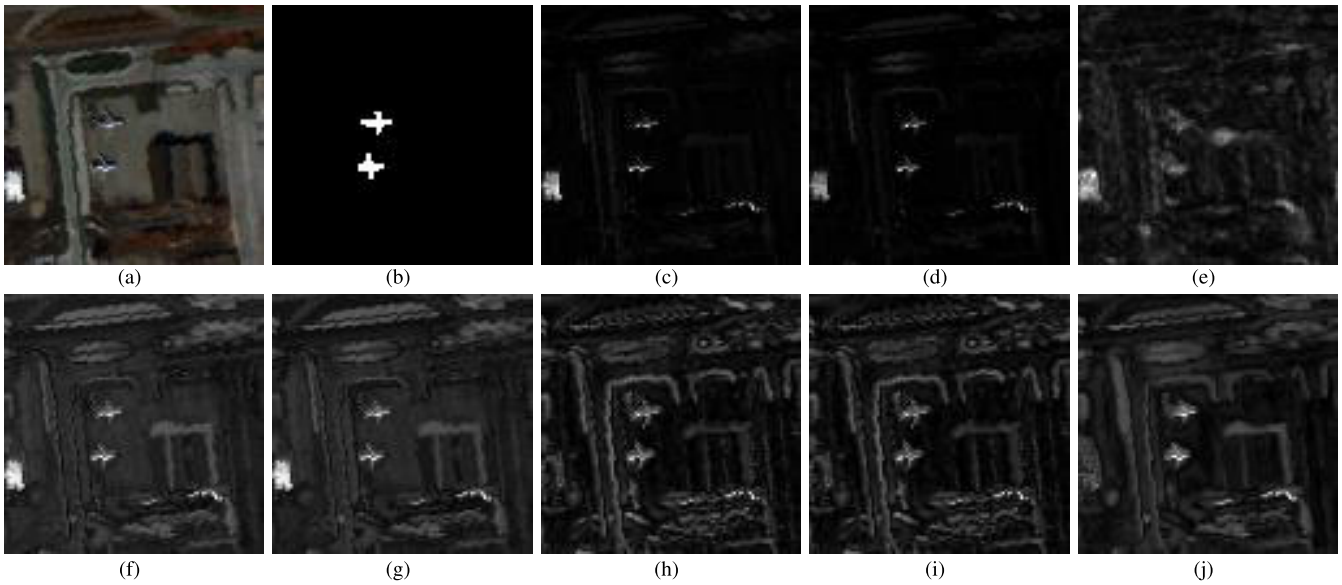


Fig. 7. Detection maps of different methods for the Airport-2 dataset. (a) False-color image. (b) Ground-truth map. (c) RX. (d) RPCA-RX. (e) TPCA. (f) LRR. (g) LRASR. (h) TVLRR. (i) GTVLRR. (j) HADGSM.

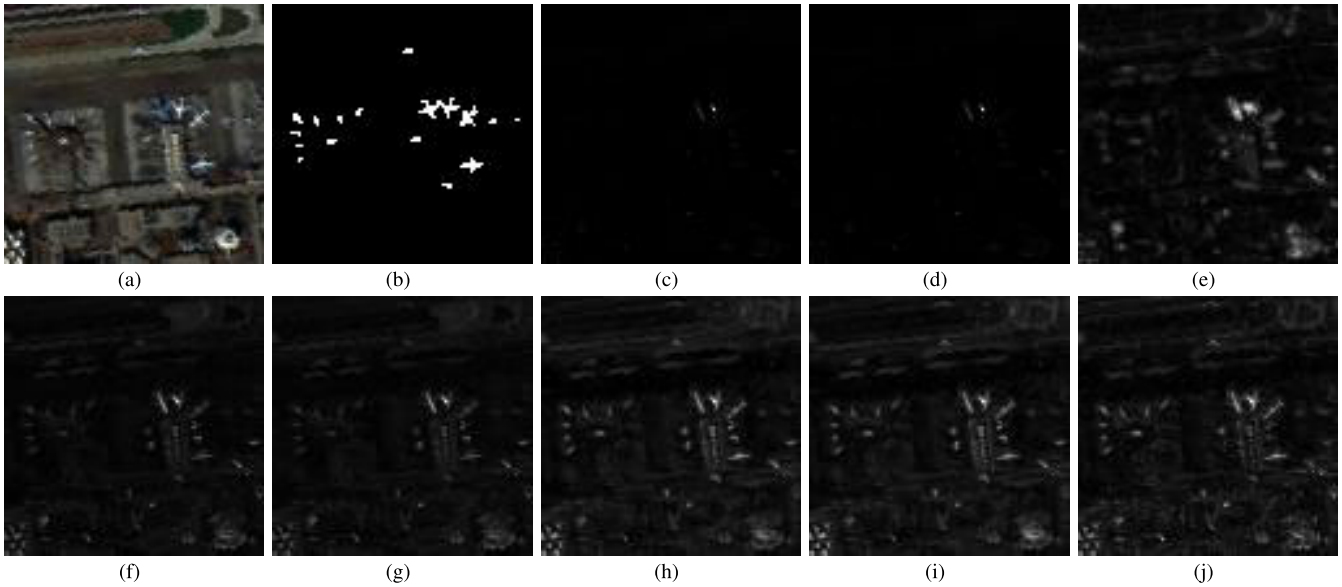


Fig. 8. Detection maps of different methods for the Airport-3 dataset. (a) False-color image. (b) Ground-truth map. (c) RX. (d) RPCA-RX. (e) TPCA. (f) LRR. (g) LRASR. (h) TVLRR. (i) GTVLRR. (j) HADGSM.

Overall, the proposed HADGSM method exhibits a higher probability of detection values than others in terms of curve comparison. Especially for Airport-4, the detection probability of the HADGSM method can reach 85% in the case of the false alarm rate being equal to 10^{-2}, while the detection probability of other methods is lower than 50%. As listed in Table III, the AUC values of the HADGSM method are all larger than those of the compared methods for all the Airport datasets. Moreover, the average AUC value of the HADGSM method is 0.02 larger than the second-highest average AUC value, which is achieved by the GTVLRR method. The obvious improvement confirms that the proposed method outperforms the classical detectors. Furthermore, the statistic-based methods have great advantages in time-consuming compared with the LRR-based methods, whereas the proposed HADGSM method has certain advantages over LRR-based methods.

The detection maps obtained by different methods for four Airport datasets are shown in Figs. 6–9. Visually, the statistic-based methods fail to detect small airplanes, e.g., the airplanes in Airport-1 and Airport-3. While for larger airplanes in Airport-2, the contours of the airplanes fail to be captured. Although the large airplane in Airport-4 is well detected, the two small airplanes are missing. The LRR-based methods perform better visualization on all the Airport datasets. At first glance, all the LRR-based methods seem to produce similar detection maps. In more detail, the LRR and LRASR methods detect the roof in the Airport-1 dataset as the anomaly and

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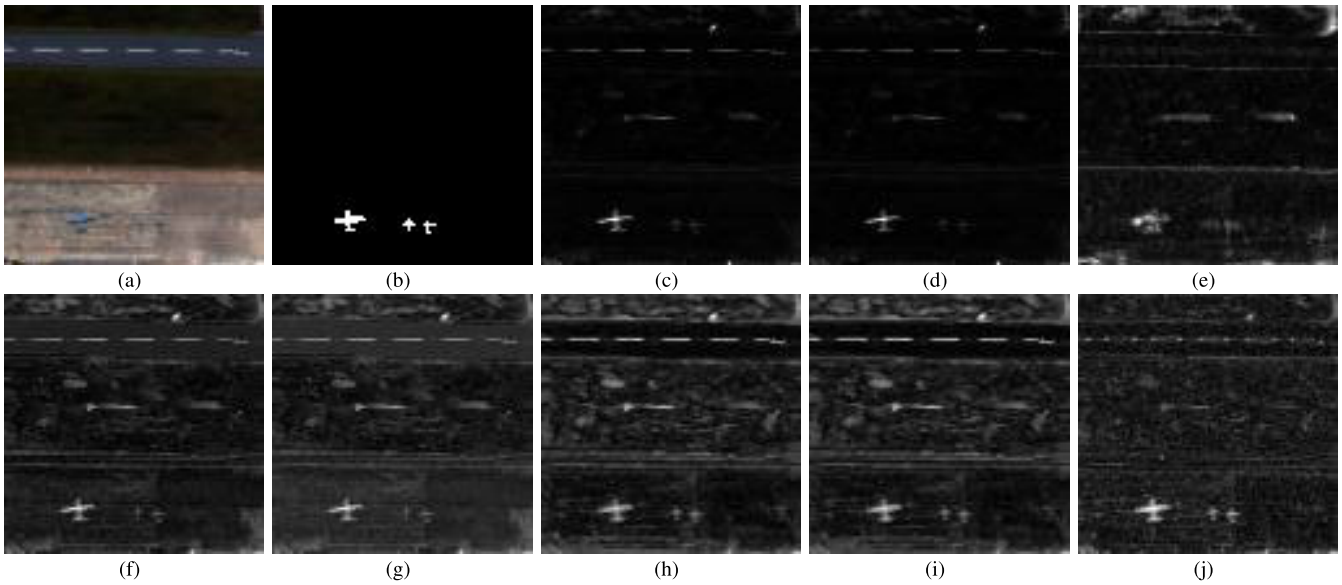


Fig. 9. Detection maps of different methods for the Airport-4 dataset. (a) False-color image. (b) Ground-truth map. (c) RX. (d) RPCA-RX. (e) TPCA. (f) LRR. (g) LRASR. (h) TVLRR. (i) GTVLRR. (j) HADGSM.

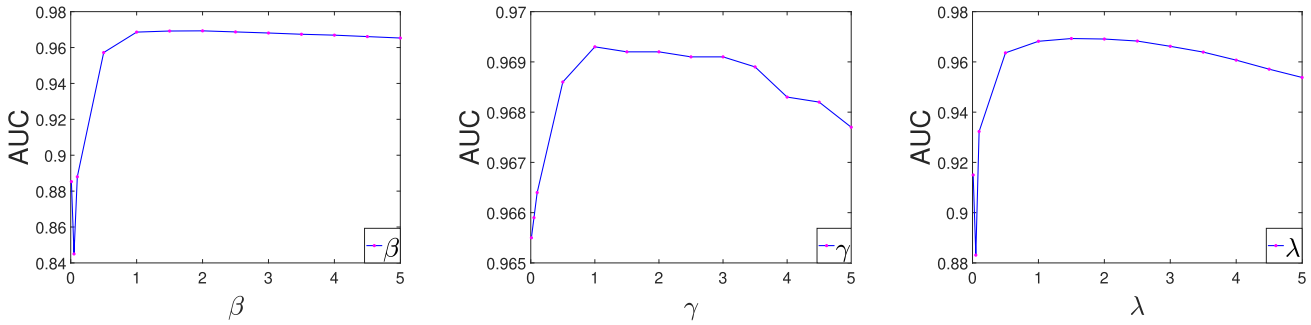


Fig. 10. Parameter sensitivity analysis for three regularization parameters (β, γ, and λ) on the Airport-2 dataset.

ignore the airplanes in the upper left corner. The detection map provided by the GTVLRR method is not very clear, while the map provided by the TVLRR method is noisy. The HADGSM method achieves the best detection performance. In Fig. 7, all the LRR-based methods detect the two airplanes. The LRR and LRASR methods detect the roof as the anomaly again, and the HADGSM model provides brighter airplane silhouettes. In Fig. 8, the detection maps provided by the LRR and LRASR methods cannot clearly show the several airplanes on the left. The TVLRR, GTVLRR, and HADGSM methods obtain much clearer detection maps. In Fig. 9, the detection maps obtained by the HADGSM method provide a clearer contour of the two small airplanes and less background interference than other methods.

D. Parameters Sensitivity Analysis

We first assess the effect of the parameter a in GSMs on detection performance. The AUC values obtained by the HADGSM method using different a values for the San Diego and Hydice-Urban datasets are listed in Table IV. The GSMs-based methods test the detection performance of parameter a ranging from $\{0.5, 0, -0.5\}$. It can be clearly observed that the AUC values are quite close under different values of a, which indicates that the method is robust to the parameter a. We fix $a = -0.5$ in all the experiments since the detection performance achieves the best results.

TABLE IV
AUC VALUES OBTAINED BY HADGSM USING DIFFERENT a VALUES FOR THE SAN DIEGO AND HYDICE-URBAN DATASETS

Data	HADGSM			
	a	$a = 0.5$	$a = 0$	$a = -0.5$
San Diego	AUC	0.9966	0.9966	0.9967
Hydice-Urban	AUC	0.9981	0.9984	0.9984

As the detection performance of the proposed method is sensitive to the regularization parameters β, γ, and λ, it is indispensable to discover a suitable range to guide the choice of the parameters. For this purpose, we adopt the grid search strategy to obtain the best performance of the HADGSM method. More specifically, the optimal value of the current parameter is evaluated while the other two are fixed until the largest AUC values are reached. The AUC values as a function of three regularization parameters on the Airport-2 dataset are

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plotted in Fig. 10. As one can see, the proposed HADGSM method achieves low AUC values when β, γ, and λ are too small. For the parameter β, the AUC values increase as β increases and then tend to steady. For the parameters γ and λ, a general trend of the AUC curves is to reach the maximum and then gradually decrease. Moreover, the parameter λ is more sensitive than the other two, and the parameter γ is the least sensitive of the three parameters. One strategy for parameter selection is to first select the more robust parameter γ, then select parameter β with a larger value, and finally fine-tune parameter λ within a wider range. These observations demonstrate the positive influence of the proper choice of the parameters on detection performance.

V. Conclusion

In this article, we exhibited experimental evidence of the potential of GSMs in solving the hyperspectral anomaly detection problem. The proposed method provides a new nonconvex manner to approximate the group sparsity, l_0 gradient, and low-rankness penalties in the classic LRR-based models. The main novelty of the proposed HADGSM framework is that it does not need the closed-form expression of the penalties, but only the associated GSMs. Extensive numerical experiments on several real hyperspectral datasets demonstrated the superiority of the proposed framework over state-of-the-art detectors.

Future work will concentrate on adapting the GSM-induced penalties to be applicable to the tensor-based anomaly detection methods. Moreover, the convergence of the proposed framework and the setting of the three regularization parameters in the model are also promising topics to further investigate.

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